

1. Telemetry Subsystem Scope

Subsystem Specification SAT-REF-77-OMEGA governs telemetry transmission rates, downlink frequency bands, and onboard packet buffering for the Leo-Sat orbital platform. The primary telemetry downlink operates on the Ka-band frequency of 14.25 GHz with a maximum bandwidth of 350 MHz.

2. Transmission Protocols and Error Codes

Under nominal conditions, packet loss must not exceed 0.001%. If orbital drift exceeds 0.05 degrees, the attitude controller raises error code ERR_ORBIT_9921, initiating an automated thruster realignment burn.

3. Fiscal Allocation and Power Consumption

Total allocated orbital budget for the SAT-REF-77-OMEGA subsystem is \$8,500,000 for fiscal cycle 2026. Nominal operating power consumption is 320 Watts during downlink and 45 Watts during solar occultation standby mode.

Subsystem Component	Power (W)	Weight (kg)	Operating Temp
Transponder Array	180 W	12.4 kg	-20C to +55C
Diplexer Unit	40 W	4.2 kg	-40C to +70C
High-Gain Antenna	100 W	18.5 kg	-50C to +85C